

MANASA SUDHARSHANAM

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PROFESSIONAL SUMMARY

- 5+ years of experience as a Data Engineer specializing in cloud-native data pipelines, ETL/ELT workflows, and data warehousing on AWS and Azure platforms.
- Proficient in designing and implementing large-scale batch and real-time data pipelines using AWS Glue, Azure Data Factory, PySpark, Redshift, Snowflake, Kinesis, and Databricks, processing multi-terabyte datasets daily.
- Expert in data modeling, analytics, and BI, building dimensional/star schemas, curated datasets, and dashboards using Power BI, Tableau, QuickSight, and Looker to support cross-functional business decisions.
- Strong background in cloud architecture and DevOps, including containerization with Docker/Kubernetes, CI/CD pipelines with Jenkins/CodePipeline, and secure, compliant data governance practices (HIPAA, PCI-DSS, SOC2).
- Proven track record of performance optimization and operational excellence, improving pipeline runtimes by up to 50%, achieving >99% reliability, and automating monitoring, alerting, and lifecycle management for large-scale enterprise datasets.

TECHNICAL SKILLS

Category	Skills
Programming Languages	Java, SQL, Typescript, Python, Bash Scripting, Go, R, C#, Scala, Ruby
Amazon Web Services (AWS)	S3, SNS, DynamoDB, Backup, EC2, Lambda, CloudWatch, Redshift, EMR, CloudTrail, ECS, CodePipeline, Kinesis, VPC, Glue, IAM, SQS, EKS, Athena, Glacier, Elastic Beanstalk, CloudFront, Route 53, API Gateway
Microsoft Azure Services	Stream Analytics, Databricks, Blob Storage, Logic Apps, Azure Data Factory, Azure SQL Database, Event Hubs, Azure SQL Data Warehouse, Synapse Analytics, SQL Server, Data Lake Storage, Azure Kubernetes Service (AKS), Cognitive Services, Azure Functions, Application Insights
Methodologies	Scrum, Agile, Waterfall, Kanban, Lean, DevOps
ETL & Data Warehousing	Redshift, Snowflake, Airflow, SSIS, Azure Data Factory, Glue, Synapse Analytics, Talend, Informatica, Matillion, dbt, HL7, FHIR
Databases	Cassandra, MySQL, PostgreSQL, MongoDB, Oracle, MariaDB, Redis, Neo4j
Big Data	Hive, Kinesis, PySpark, Apache Spark, Flink, Hadoop, Airflow, Kafka, Presto, Druid, Impala
Business Intelligence Tools	QuickSight, Tableau, PowerBI, Looker, Qlik Sense, SAP BusinessObjects, MicroStrategy
Containerization & Deployment	Docker, GitHub, Jenkins, Git, Kubernetes, CI/CD, BitBucket, Helm, Terraform, Ansible, ArgoCD

PROFESSIONAL EXPERIENCE

DISCOVER

Senior Data Engineer

McLean, VA
Jan 2025 – Present

- Engineered 50+ AWS Glue PySpark ETL/ELT pipelines ingesting 2.5+ TB of transactional and customer financial data daily, enabling regulatory reporting and fraud analytics.

- Developed batch and incremental ingestion workflows using AWS Glue, PySpark, Python, and Redshift SQL, implementing CDC patterns to improve data freshness and reduce pipeline runtimes by 40%.
- Modeled analytical datasets with star and snowflake schemas in Redshift and Snowflake, supporting fraud detection, credit risk reporting, and business intelligence for 5+ enterprise teams.
- Built real-time streaming pipelines with Kinesis Data Streams, Lambda, EMR, and Kafka, processing 500K+ financial transactions per minute, enabling sub-2-minute risk monitoring.
- Implemented data security and governance controls aligned with PCI-DSS, SOC2, and internal compliance standards, securing 10+ sensitive financial datasets with KMS encryption, IAM policies, and VPC isolation.
- Orchestrated end-to-end workflows using Apache Airflow, Step Functions, and Lambda, incorporating retries, backfills, SLA tracking, and automated failure recovery to achieve 99.8% pipeline reliability.
- Optimized storage layouts and query performance using Redshift distribution/sort keys, Snowflake clustering, S3 partitioning, and file compaction, reducing analytical query latency by 50% on multi-terabyte datasets.
- Monitored pipeline health and data quality with CloudWatch, CloudTrail, SNS, and Redshift performance metrics, detecting 95% of latency and ingestion issues proactively.
- Delivered 50+ dashboards and automated reports consumed by cross-functional teams including fraud, finance, operations, and risk, reducing manual reporting efforts by 70%.
- Documented pipeline architecture, schemas, and operational runbooks, accelerating new engineer onboarding by 30% and enabling traceable audit compliance.
- Deployed lightweight AWS Lambda functions for event-driven validation, schema checks, and S3 triggers, improving ingestion accuracy across 40+ datasets.
- Managed data lifecycle policies in Amazon S3 and Glacier, automating tiered archival and retention for 8TB+ of sensitive financial data, ensuring compliance and cost optimization.

INTUITIVE SURGICAL DataEngineer

**Bangalore, India
Aug 2021 - Dec 2023**

- Engineered 40+ Azure Data Factory pipelines for scheduled and event-driven ingestion of clinical, operational, and device-generated datasets, processing 1.5+ TB/day for analytics and reporting.
- Developed ETL workflows using Python, PySpark, and SQL in Azure Databricks to transform and aggregate large-scale healthcare datasets from HL7 and FHIR sources, reducing processing time by 50%.
- Modeled relational and dimensional data structures in Azure SQL Database and Synapse Analytics using HL7 and FHIR healthcare data standards, supporting reporting on surgical procedures, device utilization, and operational KPIs for 5+ clinical departments.
- Implemented data validation, reconciliation, and anomaly detection frameworks, ensuring 99.9% data accuracy across regulated healthcare datasets.
- Enforced HIPAA- and FDA-compliant data governance using Azure Data Lake Storage Gen2 RBAC, HL7/FHIR healthcare interoperability standards, encryption at rest and in transit, PII masking, and environment separation across dev/test/prod.
- Orchestrated secure end-to-end pipelines using Azure Data Factory, Databricks Jobs, and Logic Apps, achieving >99% operational uptime for critical healthcare reporting workflows.
- Delivered curated datasets for Power BI dashboards supporting surgical, device, and operational analytics, reducing manual reporting effort by 65%.
- Migrated 15+ legacy ETL workflows to cloud-native Azure platforms, increasing scalability, reliability, and maintainability for enterprise clinical data pipelines.
- Stored and managed structured and semi-structured data in Azure Data Lake Storage Gen2 using standardized folder hierarchies, partitioning strategies, and lifecycle policies, improving retrieval efficiency by 45%.
- Implemented role-based access control, environment separation, and Azure Key Vault integration to secure sensitive clinical data and ensure audit readiness.

- Built automated monitoring and alerting using Azure Monitor, Log Analytics, and Event Grid, detecting pipeline failures and data anomalies with 95% preemptive coverage.
- Documented data flows, transformation logic, and governance standards to support FDA audits, HIPAA compliance, and internal operational reviews.

CIGNA HEALTHCARE

Data Engineer

Bangalore, India

Mar 2020 - Aug 2021

- Engineered 60+ AWS Glue PySpark ETL/ELT pipelines processing 3+ TB of policy, claims, and customer data daily, ingesting into S3 and Redshift curated zones for analytics and regulatory reporting.
- Built real-time fraud detection pipelines using Kinesis Data Streams, EMR, Lambda, and Kafka, streaming 500K+ insurance transactions per minute to Redshift, reducing detection time by 65%.
- Optimized Redshift and Snowflake warehouses with sort/distribution keys, clustering, and materialized views, improving query performance by 50% for multi-terabyte actuarial and claims workloads.
- Developed CDC frameworks and incremental load patterns for 10+ legacy source systems, ensuring 99.95% data replication accuracy for downstream analytics and reporting pipelines.
- Automated CI/CD deployments using AWS CodePipeline, Jenkins, Git, and Lambda, managing 120+ Glue and Redshift deployments per month with rollback support, achieving 99.8% deployment success rate.
- Orchestrated multi-step workflows using AWS Step Functions and Lambda, implementing SLA-aware retries, backfills, and failure recovery, increasing pipeline throughput by 40%.
- Containerized 30+ Python-based ETL jobs using Docker, deploying on EKS clusters with VPC isolation, enabling auto-scaling and reproducible execution across dev, staging, and production.
- Built automated data quality monitoring framework with PySpark, SQL, and AWS CloudWatch, validating 50+ insurance datasets, reducing monthly pipeline errors by 70%.
- Managed S3 data lifecycle policies and implemented tiered storage (Standard, Infrequent Access, Glacier) for 8+ TB of insurance datasets, optimizing storage costs by 35% while maintaining compliance.
- Designed dimensional and star-schema models in Redshift and Snowflake for policy, claims, and underwriting data, supporting 50+ dashboards and reports consumed by actuarial, finance, and operations teams.

EDUCATION

Masters in Management Information Systems at the University of Houston, TX